

Presentation located on AFDO website (Association of Food and Drug Officials):
Chemical compounds created through a chemical process by human agency, as opposed to those of natural origin.

NIDA (National Institute Drug Abuse):
a class of lab-made substances that are chemically similar to chemicals found in the cannabis plant

EMCDDA (European Monitoring Centre for Drug and Drug Addiction):
synthetic cannabinoids' are defined as new psychoactive substances that mimic the effects of tetrahydrocannabinol, the major psychoactive substance in cannabis.
UNODC (United Nations Office on Drugs and Crime):

Synthetic cannabinoids are referred to as substances with structural features which allow binding to one of the known cannabinoid receptors, i.e. CB1 or CB2, present in human cells.

ADF (Alcohol and Drug Foundation Australia):
Synthetic cannabinoids are a large group of chemical compounds that target your body's cannabinoid receptors to produce their effects.² (If you're unsure what these receptors are – they are cell receptors located throughout your body that impact on things like how hungry you are, pain-sensation, mood, and memory).⁵ While the technically correct term for these products is 'synthetic cannabinoid receptor agonists' (SCRA), for simplicity we use the term 'synthetic cannabinoids'.

ACMD (Advisory Council on the Misuse of Drugs) – U.K.:
Any compound (not being a compound for the time being specified in sub-paragraph (c) above) structurally related to 1-pentyl-3-(1-naphthoyl)indole (JWH-018), in that the four substructures, that is to say the indole ring, the pentyl substituent, the methanone linking group and the naphthyl ring, are linked together in a similar manner, whether or not any of the sub-structures have been modified, and whether or not substituted in any of the linked sub-structures with a benzyl or phenyl group and whether or not such compound is further substituted to any extent with alkyl, alkenyl, alkoxy, halide, haloalkyl or cyano substituents and, where any of the sub-structures have been modified, the modifications of the sub-structures are limited to any of the following, that is to say—
(i) replacement of the indole ring with indane, indene, indazole, pyrrole, pyrazole, imidazole, benzimidazole, pyrrolo[2,3-b]pyridine, pyrrolo[3,2-c]pyridine or pyrazolo[3,4-b]pyridine;
(ii) replacement of the pentyl substituent with alkyl, alkenyl, benzyl, cycloalkylmethyl, cycloalkylethyl, (N-methylpiperidin-2-yl)methyl, 2-(4-morpholinyl)ethyl or (tetrahydropyran-4-yl)methyl;
(iii) replacement of the methanone linking group with an ethanone, carboxamide, carboxylate, methylene bridge or methine group;
(iv) replacement of the 1-naphthyl ring with 2-naphthyl, phenyl, benzyl, adamantyl, cycloalkyl, cycloalkylmethyl, cycloalkylethyl, bicyclo[2.2.1]heptanyl, 1,2,3,4-tetrahydronaphthyl, quinolinyl, isoquinolinyl, 1-amino-1-oxopropan-2-yl, 1-hydroxy-1-oxopropan-2-yl, piperidinyl, morpholinyl, pyrrolidinyl, tetrahydropyranyl or piperazinyl.”

NHS (National Health Service):
Synthetic cannabinoids are lab-made drugs.

CDC:
Synthetic cannabinoids (“synthetic marijuana,” “Spice,” “K2”) are various manmade chemicals that some people may use as an alternative to marijuana. These seemingly innocent little packages of “fake weed” can cause serious side effects that are very different from those of marijuana.

NDI (New Dietary Ingredient) Notification 2016 guidance (FDA-2011-D-0376):
Below are some examples of processes that FDA would likely consider to involve chemical alteration. These processes would also be likely to affect the safety profile of a dietary ingredient. The examples below are intended only for the purpose of illustration and are not a comprehensive list of processes that result in chemical alteration. See question IV.B.5 for further discussion on chemical alteration.

A process that makes or breaks chemical bonds, unless the bonds created by the process are reversed when the ingredient is dissolved in water (e.g., creation of a soluble salt) or during ingestion. Example: hydrolysis.

Removal of some components of a tincture or solution in water, which changes the chemical or molecular composition or structure of the mixture. Examples: chromatography, distillation, and filtration.

Use of solvents other than water or aqueous ethanol to make an extract or tincture. The official legislative history of DSHEA specifies that “solution in water” and “tincture” (solution in aqueous ethanol) are not processes that chemically alter a food.²³ However, other solvents typically alter the composition of the extract in significantly different ways, usually by extracting different types of constituents than are extracted using water and aqueous ethanol.

High temperature baking or cooking of an ingredient that has not previously been baked or cooked, unless the process causes only minor loss of volatile components with no other changes to the chemical or molecular composition or structure of the ingredient.

Changing the manufacturing method for an ingredient such that the chemical or molecular composition or structure is significantly different. Examples: changes that alter the composition of materials used to make the ingredient, use of a different solvent, or use of a chromatographic matrix instead of a passive filter.

Application of nanotechnology that results in new or altered chemical properties of the ingredient.

Changing agricultural or fermentation conditions to alter the chemical or molecular composition or structure of the ingredient. Examples: sprouting garlic or fermenting yeast using

a medium containing large amounts of sodium selenite to create large amounts of organic selenium compounds.

Fermentation using a fermentation medium different from the one used to make conventional foods in the food supply. Example: use of a defined commercial growth medium to produce a microorganism previously made by fermenting milk into dairy products like yogurt or cheese.

Use of a botanical ingredient that is at a different life stage than the life stage of the botanical ingredient used as a conventional food. Examples: making an extract from unripe instead of ripe apples or

Some additional thoughts concerning synthetic cannabinoids:

- Does the definition need to say something about lacing/coating/spraying/infusing of carrier material like hemp and herbs etc.?
- Some literature uses the term “Semi-synthetics” (for example Hexahydrocannabinol) for those cannabinoids that maintain the chemical structure of THC but on which small chemical modifications are produced.
- Synthetically created cannabinoids may or may not include natural occurring compounds.

Farm Bill:

“All synthetically derived tetrahydrocannabinols remain schedule I controlled substances” 85 CFR 51639S1 **Analysis Plan**

~~The objective of the present study is to examine the implications of anomalously warm weather for individuals living with multiple sclerosis (MS), a chronic demyelinating disease of the central nervous system (CNS). Our analysis leverages extant data from the Optum Clinformatics Data Mart Database (OptumInsight, Eden Prairie, MN), a large, de-identified commercial and Medicare Advantage claims database. Patient-level commercial claims data are available from January 1, 2003 to December 31, 2017. Member enrollment data; diagnostic codes from outpatient, emergency department, and inpatient visits; and pharmacy claims are deterministically linked across file types with a unique patient identifier. The primary objectives of our statistical analyses were to:~~

- ~~1. Examine the association between periods of anomalously warm weather and MS-related outpatient, emergency department, and inpatient visits among individuals with MS.~~
- ~~2. Examine heterogeneity in these associations by sex, age, season, and region.~~

~~Because heat sensitivity is a recognized clinical feature of MS, we hypothesized that risk of MS-related healthcare visits would be increased during periods of anomalously warm weather. Below, we provide details of our analysis plan. We specify which analyses were incorporated post hoc to further characterize the relationship between anomalously warm weather and MS-related healthcare visits, and which were incorporated in response to comments made during peer review.~~

Study population

Individuals eligible for the present study were between the ages of 18–64 years, resided in any of the lower 48 states or District of Columbia, were eligible for insurance for at least one month, and were identified as having MS based on a previously validated algorithm.[1] This algorithm requires at least three MS-related claims for any combination of inpatient, outpatient, or disease-modifying therapy (DMT) claims within 365 days. MS-related inpatient and outpatient claims are identified using primary diagnostic code 340 from the *International Classification of Diseases, Ninth Revision (ICD-9)*, and G35 from the *ICD-10*. Prescription claims for DMT include interferon beta-1a SC, interferon beta-1a IM, interferon beta-1b SC, pegylated interferon beta-1b, glatiramer acetate, dimethyl fumarate, fingolimod, siponimod, teriflunomide, cladribine, mitoxantrone, alemtuzumab, ocrelizumab, or natalizumab.

Exposure—Anomalously Warm Weather

The primary exposure of interest for the present study is anomalously warm weather, which will be defined using temperature data at the county-month level. Temperature data are available through the European Centre for Medium-Range Weather Forecast's (ECMWF) open-access ERA5 reanalysis.[2] We will define anomalies as a deviation of monthly average temperatures from the long-term average temperature over the study period for each calendar month by at least 1.5°C. Alternative metrics will include a continuous measure of the deviation between monthly average and long-term average temperatures, continuous, monthly average temperature and a measure of cool measure anomalies in which the monthly average temperatures is at least 1.5°C below the long-term average for that calendar month. We will also create a series of alternative exposure measured in which we varied the threshold at which we defined anomalously warm weather (0.5, 1.0, 2.0°C above the long-term average).

Modifications carried out during peer review

In response to comments made during peer review, we used the ECMWF temperature data to create several alternative exposure metrics. These included:

- A version of the main exposure variable in which the anomalies were lagged forward one month.
- A version of the main exposure variable in which the anomalies were lagged backward one month.
- A categorical variable for the deviation between monthly average and long-term average temperature, in which the categories were defined as $\geq 0.0^\circ\text{C}$, > 0.0 and $\leq 1.5^\circ\text{C}$, > 1.5 and $\leq 3.0^\circ\text{C}$, > 3.0 and $\leq 4.5^\circ\text{C}$ and $> 4.5^\circ\text{C}$.
- A continuous measure of the number of days each month where the temperature exceeded the long-term average for that calendar day.
- A categorical measure of the number of days each month where the temperature exceeded the long-term average for that calendar day, with categories defined as 1–6; 7–9; 10–13; 14 or more anomalous days with zero anomalous days as the referent groups.

Outcome—MS-related healthcare visits

We will identify MS-related outpatient, emergency department, and inpatient visits using diagnostic code 340 (*ICD-9*) and G35 (*ICD-10*) in any diagnostic position. We will then create an indicator variable for each visit type that equals one for any month with at least one MS-related outpatient, emergency, or inpatient visit—respectively—and equals zero otherwise. We will create an indicator variable for visits unrelated to MS which equals one for outpatient, emergency, or inpatient visit in which the diagnostic codes 340 and G35 are absent and zero otherwise.

Modifications carried out during peer review

We modified our outcome definition to include only MS-related healthcare visits to only the first, second, and third diagnostic position.

Covariates

We will define sex (women, men), categorical age (18–25; 26–35; 36–45; 46–55; 56–64 years), and calendar year using eligibility files. We will define time-varying county, state, and region of residence using five-digit zip codes and season for each respondent.

Modifications carried out during peer review

In response to requests from reviewers, we defined age as a continuous rather than categorical variable in the analyses described below, and assigned climate zones based on county of residence using data from the U.S. Department of Energy.[3]

Statistical Analysis

We will conduct an individual-level, repeated measures analysis. For our primary analysis, we will use generalized linear models with the binomial family and log link to estimate risk ratios (RR) for the association between anomalously warm weather and MS-related outpatient, emergency department, and inpatient visits, respectively. We will adjust for *a priori* specified covariates that could potentially confound the association of interest. These include sex, categorical age, and fixed effects for state and calendar year. For all analyses, we will calculate robust standard errors to account for repeated measures within individuals and potential non-independence of outcomes within counties. We will use fitted models to estimate associations on the additive scale. Because of sex differences in MS incidence, response to treatment, and healthcare utilization,[4–8] we repeated our primary analyses stratified by sex.

Secondary analyses will be conducted within subgroups defined by age category, region, season, and by region within season. We will also examine the association between anomalously cool weather and MS-related healthcare visits. Sensitivity analyses will include an examination of the association between the continuous difference between monthly average and long-term average temperatures and MS-related healthcare visits; the association between average monthly temperature and MS-related healthcare visits; and an analysis using alternative thresholds to define anomalies (0.5, 1.0, 2.0°C).

All statistical analyses were performed with R version 3.2.3 (R Foundation for Statistical Computing, Vienna, Austria).

Modifications carried out during peer review

We made several revisions to our statistical analyses in response to comments made during peer review, which are detailed below:

- We defined age as continuous with natural splines rather than as a categorical variable in all models.
- Secondary analyses
 - We included a subgroup analysis by climate zone.
- Formal test of interaction
 - We conducted formal tests of interaction using a p-value for interaction between our exposure variable and sex, age group, season, region, and climate zone.
- Sensitivity analyses
 - We further characterized the importance of the magnitude of the deviation of monthly average temperatures from long-term average temperatures by (1) repeating our main analysis using a categorical variable for the difference between monthly average and long-term average temperatures, and (2) defining an alternative, categorical metric for the number of anomalously warm days per month as a categorical variable.
 - We lagged the exposure metric forward and backward by one month and then repeated our main analysis.

- ~~Negative exposure controls~~

- ~~Because our negative outcome control analysis still showed persistent—albeit attenuated—associations, we included a new negative exposure controls analysis in which we repeated our main analysis after randomly permuting the exposure variable.~~
- ~~Because cold sensitivity is not a clinical feature of multiple sclerosis, we feel that our analysis of anomalously cool weather—which we initially presented as a secondary analysis—is more appropriate as a negative exposure control analysis.~~

References

1. ~~Culpepper, W.J., et al., *Validation of an algorithm for identifying MS cases in administrative health claims datasets*. Neurology, 2019. **92**(10): p. e1016–e1028.~~
2. ~~ECMWF. *ERA5*. 2020 [cited 2020 April 10]; Available from: <https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>.~~
3. ~~Baechler, M.C., et al., *Guide to determining climate regions by county*. Pacific Northwest National Laboratory & Oak Ridge National Laboratory, 2010. **7**: p. 1–34.~~
4. ~~Koch-Henriksen, N., et al., *Incidence of MS has increased markedly over six decades in Denmark particularly with late onset and in women*. Neurology, 2018. **90**(22): p. e1954–e1963.~~
5. ~~Alonso, A. and M.A. Hernán, *Temporal trends in the incidence of multiple sclerosis: a systematic review*. Neurology, 2008. **71**(2): p. 129–135.~~
6. ~~Magyari, M., et al., *Gender effects on treatment response to interferon-beta in multiple sclerosis*. Acta Neurologica Scandinavica, 2014. **130**(6): p. 374–379.~~
7. ~~Manteuffel, M., et al., *Influence of patient sex and gender on medication use, adherence, and prescribing alignment with guidelines*. Journal of women's health, 2014. **23**(2): p. 112–119.~~
8. ~~Owens, G., *Gender differences in health care expenditures, resource utilization, and quality of care*. Journal of Managed Care Pharmacy, 2008. **14**(3): p. 2–6.~~

